A Behavioral Comparison of Adversarially Trained ResNet-18 and ViT-Tiny on CIFAR-10: Transfer Metrics, Gradient Structure, and Block-wise Feature Drift

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***Abstract*—How convolutional neural networks (CNNs) and Vision Transformers (ViTs) differ under adversarial attack is usually summarized by a single robust accuracy number. This paper reports preliminary results from an ongoing behavioral comparison of adversarially trained (AT) ResNet-18 and ViT-Tiny on CIFAR-10, evaluated with AutoAttack on the full test set at $\epsilon = 8/255$ over three independent training runs per architecture, and argues that the single number hides most of the story. The CNN leads in absolute robust accuracy (37.93±0.14% vs 29.14±0.40%; McNemar $p < 10^{-84}$ per run), and a conditional decomposition attributes that lead to two separate factors, a 12.3-point clean accuracy advantage and a lower conditional fooling rate (48.6% vs 55.5% under PGD-10). The decomposition is not a formality: on an earlier pair of checkpoints, selected with a validation split that clean pre-training had already seen, the same pipeline reported matched conditional sensitivity and an AutoAttack ordering that slightly favored the ViT, so which factor carries the headline can change between training runs even when the absolute ordering does not. The second finding is that cross-architecture transfer asymmetry depends strongly on the conditioning protocol used to measure it: on identical models and data the measured asymmetry ranges from +4.4 to +19.4 points across the unconditioned, target-correct, both-correct, and successful-source protocols, a 4.4-fold spread, and the protocol decides whether an equivalence claim is accepted or rejected. The direction, CNN→ViT stronger, is stable across protocols and seeds. The third finding concerns gradient structure: scale-invariant measures show CNN input gradients to be sparser (Hoyer 0.493 vs 0.456), while ViT cross-sample gradient alignment is higher in absolute cosine (0.056 vs 0.038) even though the signed cosine is near zero for both models. Finally, under attack ViT features drift with an early drop and mid-network plateau profile, whereas the CNN profile degrades monotonically with depth. These are preliminary results on one dataset and one model pair; an extended analysis is in preparation.**

***Index Terms*—adversarial examples, adversarial training, CIFAR-10, convolutional neural networks, robustness, transfer attacks, Vision Transformer**

I. Introduction

Deep image classifiers are vulnerable to adversarial examples, small and carefully crafted perturbations that flip predictions with high confidence [1], [2]. As Vision Transformers (ViTs) [3] join convolutional neural networks (CNNs) [4] as a dominant architecture family, a natural question is how the two families differ under attack. Much of the literature approaches this as a contest and reports which architecture attains higher robust accuracy [5]–[7]. This paper takes a different, behavioral view: for a fixed pair of adversarially trained models, *how* do the architectures differ in the composition of the headline robustness number, in how adversarial examples transfer between them, in the structure of their input gradients, and in how perturbations propagate through their layers?

We report preliminary results from an ongoing study of this question on CIFAR-10 [8]: a ResNet-18 and a ViT-Tiny trained with the same adversarial objective and threat model, evaluated with AutoAttack [9] on the full test set. Our contributions are: (i) a conditional decomposition of robustness showing which factor, clean accuracy or conditional sensitivity, carries the "CNN is more robust" headline, and showing that this attribution can change between training runs; (ii) a multi-protocol analysis showing that the measured size of cross-architecture transfer asymmetry varies several-fold with the conditioning protocol, which serves as a methodological caution for transfer studies; (iii) paired, scale-invariant gradient structure comparisons including a signed versus absolute alignment distinction; and (iv) a block-wise feature drift comparison that reveals qualitatively different depth profiles. All results derive from per-sample-logged evaluations on the full test set, over three independent training runs per architecture. Fig. 1 summarizes the study design.

II. Related Work

Adversarial examples and gradient-based attacks were established by Szegedy et al. [1] and Goodfellow et al. [2]; projected gradient descent (PGD) adversarial training (AT) [10] remains the standard defense, with robust overfitting and early stopping characterized by Rice et al. [11]. AutoAttack [9] and RobustBench [12] standardized robustness evaluation.

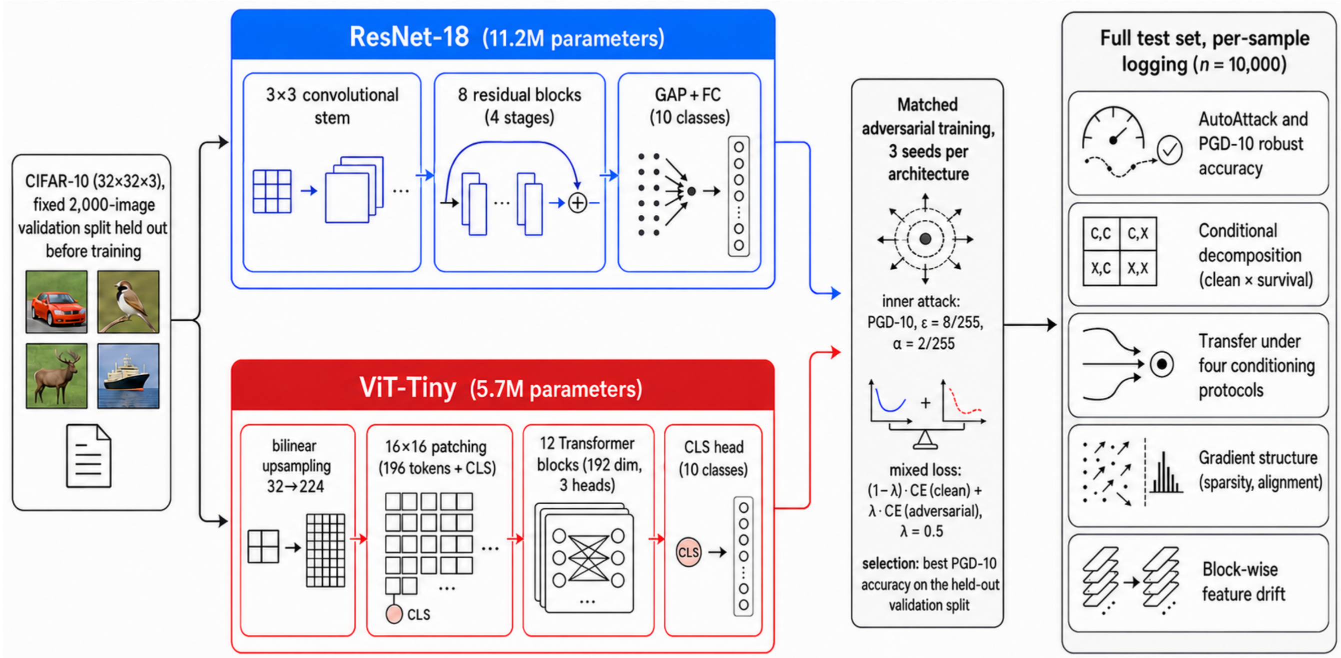

Fig. 1. Study overview: the matched adversarial training pipeline for the two architectures and the per-sample-logged evaluation battery.

Among direct CNN–ViT robustness comparisons, findings vary with the setup: Bai et al. [5] found that under fair training recipes transformers do not surpass CNNs in adversarial robustness, whereas Benz et al. [6] reported ViTs to be more robust than CNNs for standardly trained models. This contrast is consistent with our thesis that comparison outcomes are sensitive to setup and protocol. Closest to our setting, Ali et al. [7] examined the robustness of standardly trained ViTs and CNNs through FGSM (Fast Gradient Sign Method), PGD, and DeepFool score comparisons. Our study differs in three ways: both models are *adversarially trained* under a matched objective; evaluation uses AutoAttack on the full test set with paired per-sample statistics; and the comparison goes beyond scores to conditional decompositions, protocol-explicit transfer, gradient structure, and block-wise drift. ViT adversarial training is known to be recipe-sensitive on small datasets [13], [14], and with synthetic extra data generated by diffusion models ViTs can surpass WideResNets on CIFAR-10 [15]. Our conclusions therefore hold under the baseline recipe we study, not for optimally trained ViTs.

For transfer evaluation, conditioning on correctly classified samples is established practice [16]–[19]; both-correct variants were used by Mahmood et al. [18] and TREND [19]. Our contribution is to show on a single dataset that the *choice* among these protocols qualitatively changes the asymmetry conclusion.

III. Method

A. Dataset, Models, and Training

Experiments are conducted on CIFAR-10 [8]: 60,000 color images of size 32×32 in ten balanced classes (50,000 train / 10,000 test). Fig. 2 illustrates the dataset and the effect of the attack. The example shown is the first sample in test order that is correctly classified on clean input and whose class the attack flips, a deterministic selection criterion; the perturbations are hard to notice at native scale yet change predictions with high confidence, and they are qualitatively different in structure across the two architectures (diffuse for the CNN, patch-aligned for the ViT). Images are 32×32 and are displayed with pixel-preserving (nearest neighbor) magnification. We compare a ResNet-18 (11.2M parameters; CIFAR-style stem) with a ViT-Tiny (5.7M parameters; `timm` [20] architecture, 16×16 patches over inputs bilinearly upsampled to 224×224); both models are trained from scratch on CIFAR-10 with basic data augmentation and then adversarially fine-tuned. All operations including attacks are defined in the native 32×32 CIFAR-10 pixel space: the $32 \to 224$ bilinear upsampling is a differentiable layer *inside* the ViT wrapper (`nn.Upsample`). The two models therefore share the same L_∞ threat model ($\epsilon = 8/255$ in raw $[0, 1]$ pixel space), and transferred adversarial examples are fed to both models as the same 32×32 inputs.

Adversarial training minimizes the mixed objective $(1 - \lambda)\mathcal{L}_{CE}(f(x), y) + \lambda \mathcal{L}_{CE}(f(x_{adv}), y)$ with $\lambda = 0.5$, where x_{adv} is generated in training mode with PGD-10 ($\epsilon = 8/255$, $\alpha = 2/255$, random start from $\mathcal{U}(-\epsilon, \epsilon)$, single restart) following standard practice [10], [11]. Clean pretraining runs for 200 epochs (CNN: SGD, learning rate 0.1, weight decay 5×10^{-4}; ViT: AdamW [21], 10^{-3}, decay 0.05; cosine annealing); adversarial fine-tuning runs for at most 100 epochs with learning rate 10^{-3} and batch size 128 (CNN) / 64 (ViT). Data augmentation is random cropping with 4-pixel

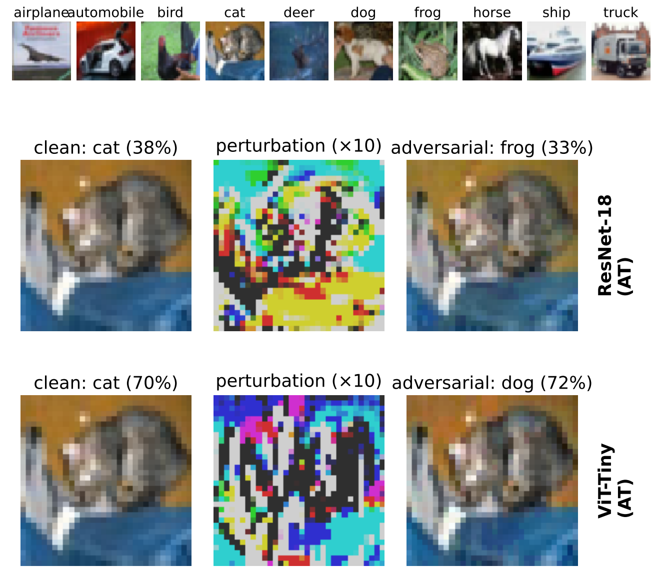

Fig. 2. Top: one sample from each CIFAR-10 class. Bottom: each model's own white-box PGD-10 attack ($\epsilon = 8/255$) on the same test image, with the clean prediction, the perturbation ($10\times$), and the adversarial prediction.

padding and horizontal flipping. Early stopping and model selection are driven by PGD-10 accuracy on a fixed 2,000-sample validation split (patience 20, threshold 0.1 points); that split is drawn once and excluded from the training set of *both* the clean and the adversarial stage, so no model is selected on data it has been trained on. The test set is touched only for final evaluations. The full pipeline is repeated three times per architecture with independent seeds (CNN 1001–1003, ViT 2001–2003), and every table reports mean±standard deviation over those runs. We also report, where it changes a conclusion, the corresponding value from an earlier single run in which the validation split had been part of clean pre-training; the contrast is used deliberately, as a measurement of how much a finding can move between training runs. As an upper reference we include the WideResNet-28-10 checkpoint of Gowal et al. [22] distributed through RobustBench [12]; since that model is trained with extra data and a stronger recipe, it is kept visually and analytically separate from the matched pair.

B. Evaluation and Metrics

All evaluations are run on the full 10,000-sample test set with per-sample outcome logging. Robustness is measured at $\epsilon = 8/255$ with FGSM, PGD-10, and the standard AutoAttack (AA) ensemble (official implementation, version 0.1: APGD-CE, APGD-T, FAB-T, Square). Alongside absolute robust accuracy we report the *conditional fooling rate*, that is, how many of the samples classified correctly on clean input the attack overturns. Since samples misclassified on clean input count as non-robust, absolute robustness factors exactly as robust $=$ clean $\times$ $(1 -$ conditional fooling), separating the clean accuracy component from the conditional sensitivity component. We further evaluate the *both-correct* subset, the samples both models classify correctly on clean input, which enables one-to-one paired comparisons.

For transfer, adversarial examples generated on a source model are evaluated on the other (target) model under three conditioning protocols: *target-correct* (the target classifies the clean sample correctly), *both-correct* [18], [19], and *successful-source* (the target is correct and the attack succeeds in the white-box sense on the source). Confidence intervals (CIs) are given by percentile bootstrap (10,000 resamples). Under the both-correct protocol the fooling indicators of the two directions are defined on the *same* samples: the paired bootstrap resamples the members of the joint set, and the sign-flip permutation test is built on per-sample directional differences. The main ± 2 point margin of the two one-sided equivalence tests (TOST) was chosen on the principle of treating as practically equivalent any difference below the variation observed at the training-run level (a few points); the ± 1 and ± 3 sensitivities are also reported in the text.

Gradient structure is examined through per-sample loss gradients on 500 fixed test samples: scale-invariant sparsity (Hoyer, Gini [23], and the fraction of components below 1% of the per-sample maximum) with per-sample paired Wilcoxon tests (Holm-corrected); cross-sample gradient alignment is reported as the mean absolute cosine over all sample pairs, together with the signed mean. Block-wise feature drift is measured on the first 100 test samples (unselected, regardless of correctness) using each model's *own* white-box PGD-10 examples: for the ViT, the MLP sub-block output of each transformer block is flattened per sample over all tokens (CLS + 196 patches); for the CNN, each residual block output is flattened over the spatial dimensions. We then compute the cosine similarity and the norm change between clean and adversarial representations (12 blocks for the ViT, 8 residual blocks for the CNN).

IV. Results

A. Robustness: One Headline, Three Readings

Table I and Fig. 3 give the absolute results: the adversarially trained CNN leads the ViT under every attack (AutoAttack 37.93±0.14% vs 29.14±0.40%; on joint samples McNemar $p < 10^{-84}$ in each of the three runs), and both trail the reference WideResNet, whose advantage is the joint product of capacity, extra data, and a stronger recipe [22]. Fig. 4 shows the gap to be stable across perturbation budgets.

The conditional decomposition (Table II) separates the two ingredients of that headline. Because misclassified clean inputs count as non-robust, absolute robustness factors exactly as robust $=$ clean $\times (1 -$conditional fooling); for AutoAttack this reads $85.78 \times 0.442 = 37.93$ for the CNN and $73.53 \times 0.396 = 29.14$ for the ViT. Both factors favor the CNN: it starts from a 12.3-point higher clean accuracy *and* loses a smaller fraction of what it solves (conditional fooling 48.58% vs 55.53% under PGD-10, 55.78% vs 60.37% under AutoAttack). On the 7,061 jointly solved samples, which give a one-to-one paired comparison, the CNN leads by 13.8 and 10.7 points

TABLE I
ROBUSTNESS ON THE FULL CIFAR-10 TEST SET ($\epsilon = 8/255$); MEAN±STD OVER THREE INDEPENDENT TRAINING RUNS PER ARCHITECTURE.

Model	AT	Clean	FGSM	PGD-10	AA
ResNet-18	–	95.25±0.15	40.29±1.77	0.03±0.03	–
ViT-Tiny	–	80.09±0.60	6.10±1.47	0.05±0.05	–
ResNet-18	✓	**85.78**±0.36	**53.46**±0.08	**44.11**±0.50	**37.93**±0.14
ViT-Tiny	✓	73.53±0.55	36.31±0.42	32.69±0.22	29.14±0.40
WRN-28-10†	✓	89.48	70.91	66.92	62.76

†External reference checkpoint [22]: stronger recipe and extra data, single checkpoint; the AA value is as reported by RobustBench.

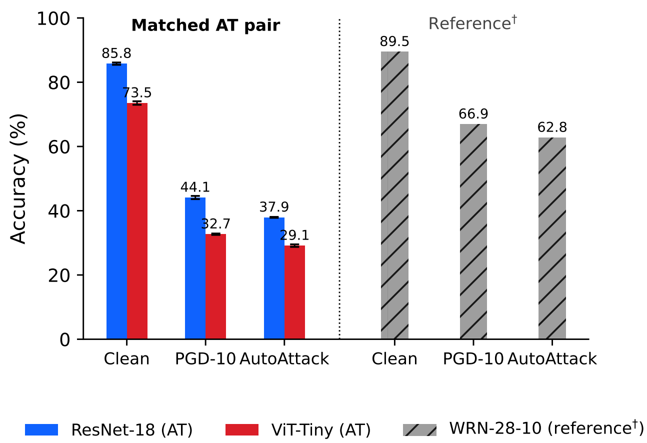

Fig. 3. Clean, PGD-10, and AutoAttack accuracies. WRN-28-10 (hatched) is an external reference trained with a stronger recipe and extra data.

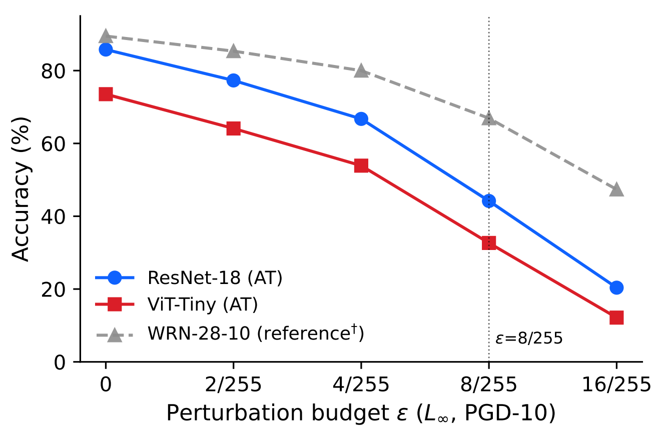

Fig. 4. PGD-10 accuracy as a function of the perturbation budget (full test set).

(McNemar $p < 10^{-84}$ in every seed). Seed-to-seed variation is small throughout ($\leq$0.8 points), so these orderings are not run-to-run noise.

The decomposition is worth reporting even when the two factors agree, because they need not. Our first checkpoints, selected with a validation split that clean pre-training had already seen, produced the same absolute ordering but the opposite conditional one: fooling rates that were indistinguishable under

TABLE II
CONDITIONAL DECOMPOSITION (FULL TEST; MEAN±STD OVER THREE SEEDS). BOTH-CORRECT: THE $n = 7{,}061 \pm 68$ SAMPLES BOTH MODELS CLASSIFY CORRECTLY ON CLEAN INPUT.

Model	Conditional fooling (%) PGD-10	Conditional fooling (%) AA	Both-correct robust (%) PGD-10	Both-correct robust (%) AA
ResNet-18 AT	**48.58**±0.80	**55.78**±0.23	**59.86**±0.35	**51.89**±0.49
ViT-Tiny AT	55.53±0.50	60.37±0.33	46.11±0.59	41.15±0.39
Leak-affected single run (same pipeline, earlier checkpoints):				
ResNet-18 AT‡	52.15	58.16	54.92	48.15
ViT-Tiny AT‡	52.33	**56.46**	49.61	45.33

PGD-10 (52.15% vs 52.33%) and slightly favored the ViT under AutoAttack (58.16% vs 56.46%), which supported the reading that the CNN advantage was entirely a clean-accuracy effect (lower block of Table II). None of the three corrected runs reproduces that reading, and they agree with each other far more closely than any of them agrees with the earlier run. The lesson is not that one number is right and the other wrong, but that a single robust accuracy hides which factor is doing the work, and that the factor attribution can change between training runs even when the absolute ordering does not.

B. Transfer: The Protocol Is Part of the Result

Table III reports cross-architecture transfer under the unconditioned (raw) rate and three established conditioning protocols, with the denominator of each protocol given in the table footnote. On identical models and identical data, the measured asymmetry ranges from +4.4 to +19.4 points depending only on which protocol is used: the spread between the largest and the smallest protocol estimate is 15.0±0.8 points, a factor of 4.4. The raw rates exaggerate the asymmetry because the ViT target carries a twelve-point higher clean error, which enters the numerator as a target weakness rather than as transfer. The target-correct protocol removes that confounder (+4.36±0.44), and the both-correct protocol, which scores both directions on the same samples, gives +8.27±0.24 (paired bootstrap CI [7.33; 9.22]; sign-flip permutation $p < 5 \times 10^{-5}$ and TOST equivalence rejected at ±3 points in every seed). The direction survives a change of attack: under the momentum-based MI-FGSM [17] the target-correct rates are 19.07±0.33% and 15.42±0.52%, a +3.65-point difference in the same direction. The mechanism is compositional: samples that the target solves on clean input but the source does not are three to four times more fragile than jointly solved ones (41.2±0.8% and 59.1±1.7% fooling versus 10.0% and 18.3%), and these marginal subsets differ five-fold in size (1,517±102 versus 291±16), which pulls the two target-correct rates toward each other.

How much this matters is visible in the checkpoints we analyzed first, which were selected with a validation split that clean pre-training had already seen: there the same pipeline reported target-correct *parity* (+0.63 points, TOST-equivalent at ±2) and a both-correct difference of only +5.33. The direction of the effect is therefore stable across both checkpoint sets, but its magnitude and its statistical verdict are not. The

TABLE III
TRANSFER ASYMMETRY UNDER THE UNCONDITIONED RATE AND THREE CONDITIONING PROTOCOLS (PGD-10, $\epsilon = 8/255$, FULL TEST; MEAN±STD OVER THREE SEEDS). LAST COLUMN: THE SAME QUANTITY ON THE EARLIER, LEAK-AFFECTED CHECKPOINTS.

Protocol	CNN→ViT	ViT→CNN	Diff.	Diff.[‡]
Unconditioned (raw)	41.02±0.55	27.45±0.27	+13.57	+8.27
Target-correct	19.87±0.18	15.51±0.59	+4.36	+0.63
Both-correct	18.25±0.17	9.98±0.31	+8.27	+5.33
Successful-source	36.39±0.76	17.02±0.52	+19.37	+11.17

Denominators (CNN→ViT / ViT→CNN): unconditioned $N = 10{,}000/10{,}000$; target-correct $N = 7{,}353/8{,}579$; both-correct $N = 7{,}061$ (joint set); successful-source $N = 2{,}831/3{,}814$. [‡]Single run, leak-affected selection.

practical lesson for cross-architecture transfer studies is that the conditioning protocol changes the measured asymmetry several-fold and can decide whether an equivalence claim is accepted or rejected; it should be stated explicitly and, where possible, varied.

C. Gradient Structure

On the 500 fixed test samples measured for both models, CNN input gradients are consistently but moderately sparser than those of the ViT under all three scale-invariant measures (Hoyer 0.493±0.012 vs 0.456±0.006; Gini 0.650±0.009 vs 0.610±0.005; components below 1% of the per-sample maximum 34.8±2.1% vs 26.8±0.9%), and the difference is significant under paired Wilcoxon tests in every run (Holm-corrected $p < 10^{-11}$ throughout; paired Cohen's d of 0.32–1.23). Adversarial training is what creates this ordering: in the cleanly trained counterparts of the same runs the sparser model is the ViT (Hoyer 0.380±0.009 vs 0.347±0.002), which is consistent with the strong sparsifying effect of adversarial training on CNN gradients [24]. Cross-sample gradient alignment, as mean absolute cosine over all $500 \times 499/2$ pairs, is higher for the ViT (0.057±0.001 vs 0.038±0.001); the cleanly trained pair already shows this gap (0.058 vs 0.029), so it is a property of the architecture rather than of adversarial training. The *signed* mean cosine, however, is near zero for both models (≤0.0014): what the samples share is axes of sensitivity, not consistent directions. All four quantities reproduce the ordering we measured on the earlier checkpoints to within their standard deviations, so unlike the conditional decomposition, the gradient structure findings do not depend on which checkpoint set is used.

D. Block-wise Feature Drift

Fig. 5 compares how far the representation of an attacked input departs from its clean counterpart at each block. The two architectures differ less in whether damage accumulates than in *where* it accumulates. In the ViT the drift is gradual and shallow: cosine similarity declines steadily to a minimum of 0.965±0.002 around block 8 and then recovers to 0.989 by block 12, while feature norms stay within 0.7% of their clean values throughout, so the perturbation corrupts the *direction* of the representation and not its magnitude. In the CNN the damage is concentrated: the first three stages lose little

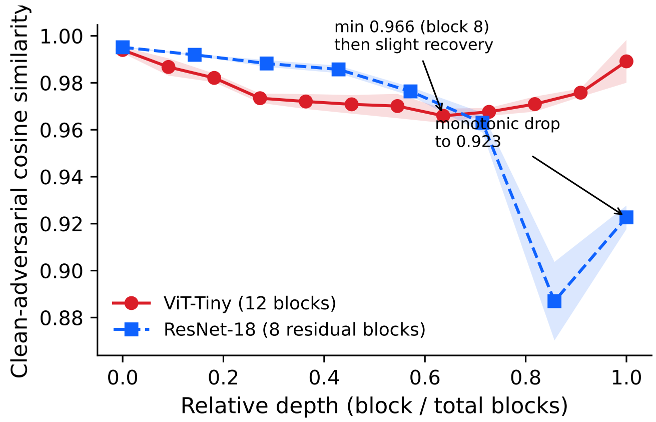

Fig. 5. Block-wise cosine similarity between clean and adversarial representations as a function of relative depth ($n = 100$; bands are ±1 standard deviation).

(0.995 to 0.963), then the first block of the last residual stage collapses to 0.887±0.017 together with a −12.9±2.6% norm contraction, after which the final block partially repairs the representation (0.923±0.005). Adversarial damage in the CNN is thus a late, localized, magnitude-changing event, whereas in the ViT it is a distributed, direction-only rotation. We note that on the earlier checkpoints the CNN profile fell monotonically to the last block; the late recovery is visible only in the corrected runs, so the shape of the CNN curve, unlike the ViT profile, should be read as recipe-dependent. The analysis carries layer-wise representation studies of cleanly trained ViTs [25] over to the adversarially trained setting. The shaded bands in Fig. 5 show ±1 standard deviation across the three runs.

V. DISCUSSION AND LIMITATIONS

Three main implications stand out. First, single-number robustness comparisons are lossy: the same headline can rest on a clean accuracy advantage, on lower conditional sensitivity, or on both, and only the decomposition says which. That the attribution moved when we changed nothing but the validation protocol is the sharpest evidence we have for reporting it. Second, claims of transfer asymmetry are relative to the protocol: measured on the same models and data, the asymmetry varied by a factor of 4.4 across four standard protocols, so cross-architecture studies should report the conditioning protocol as part of the result rather than as a footnote. The final implication is a behavioral dissociation: the architectures differ in gradient concentration and in where adversarial damage accumulates along depth, which suggests stabilization targeted at different locations for the two architectures.

Several limitations frame these results. Three runs per architecture bound seed variation but do not span the recipe space, and ViT robustness is known to be recipe-sensitive on small datasets [13]–[15]; our conclusions hold for the baseline recipe we study, not for optimally trained ViTs. The sample-level tests (McNemar, Wilcoxon, bootstrap, permutation) infer over test samples and are conditional on the trained checkpoints, so seed uncertainty enters only through the reported standard deviations. The study covers a single dataset, a single model pair, and a single-budget L_∞ threat model for the conditional analyses. The gradient and drift measurements are behavioral correlations, not established causal explanations. Finally, we cannot isolate a single cause for the differences between the two checkpoint sets. The corrected protocol both removed the selection leak and withheld 2,000 images from clean pre-training, and the two lineages are separate training runs: the earlier ViT reached 77.50% clean accuracy before adversarial training, whereas the three corrected runs reach 80.09±0.60% with *less* data, so a simple data-quantity explanation does not hold. What the comparison does establish is that conclusions of this kind are not safe to draw from one training run, since three fresh runs agree with each other ($\leq$0.8-point spread) and disagree with the single earlier one. Attributing the shift to a specific mechanism requires a controlled ablation, which we leave to the extended version along with additional transfer attacks and spatial gradient statistics.

VI. Conclusion

For a matched adversarially trained ResNet-18 / ViT-Tiny pair on CIFAR-10, measured over three independent runs per architecture, we showed that the robustness headline decomposes into a clean accuracy factor and a conditional sensitivity factor whose relative weight depends on how checkpoints are selected, that the magnitude of the measured transfer asymmetry is largely a property of the conditioning protocol while its direction is not, that gradient structure differences are consistent but moderate (with the alignment difference remaining unsigned by nature), and that adversarial damage accumulates at qualitatively different depths in the two architectures. We offer these preliminary results as a template for protocol-explicit, multi-view robustness comparisons across architecture families.

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